FIT PM12
Hardware test dossier

LEGACY-SN1803-UNBOUND

HISTORICAL — full dossier

Generated: 2026-11-08T08:48:05.605625+00:00

INCOMPLETE / NOT RELEASED

Historical multi-date engineering evidence. Factory DNA was not recorded during these measurements. This is not a new batch acceptance certificate.

Prepared by _____

Reviewed by _____

Disposition / deviations _____

Signature and date _____

Contents

1 Scope, identity and disposition **2**

2 Bench inventory and wiring **3**

3 Test coverage and acceptance rules **6**

4 Test procedure and criteria **7**

4.1 Hardware identity and PGOOD . . . 7
4.2 JTAG chain and168 static CPLD–FPGA links . . . 7
4.3 GTX internal PRBS, MAC loopback and external link . . . 7
4.4 I2C mux address inventory and bus topology . . . 7
4.5 FPGA configuration flash ownership and read-only access . . . 7
4.6 Power sequencing and timeout shutdown . . . 7
4.7 CPU peripherals, SPI walking patterns and power telemetry . . . 7
4.8 HDMI, all digital ADC paths and stimulated TDC . . . 8
4.9 Gate delay versus internal DAC-A . . . 8
4.10 ADC differential transfer and adjacent data-bit coverage . . . 8
4.11 CFD analog monitor DC levels . . . 8
4.12 Internal DAC B/C/D transfer curves . . . 8

5 Digital link and peripheral evidence **9**

5.1 MCU / SPI . . . 9
5.2 Flash ownership . . . 9
5.3 Ethernet / PRBS . . . 9
5.4 I2C topology . . . 10
5.5 Power rail observations . . . 10

6 CFD and ADC — channel comparison **12**

7 TDC phase response **16**

8 Internal CFD DAC B/C/D characteristics **17**

9 Static connection results **18**

10 Raw data, reproducibility and review **21**

1 Scope, identity and disposition

This dossier records evidence and test coverage. A long report is not proof of compliance: incomplete channels, unknown configuration, absent raw data and failed restoration remain visible.

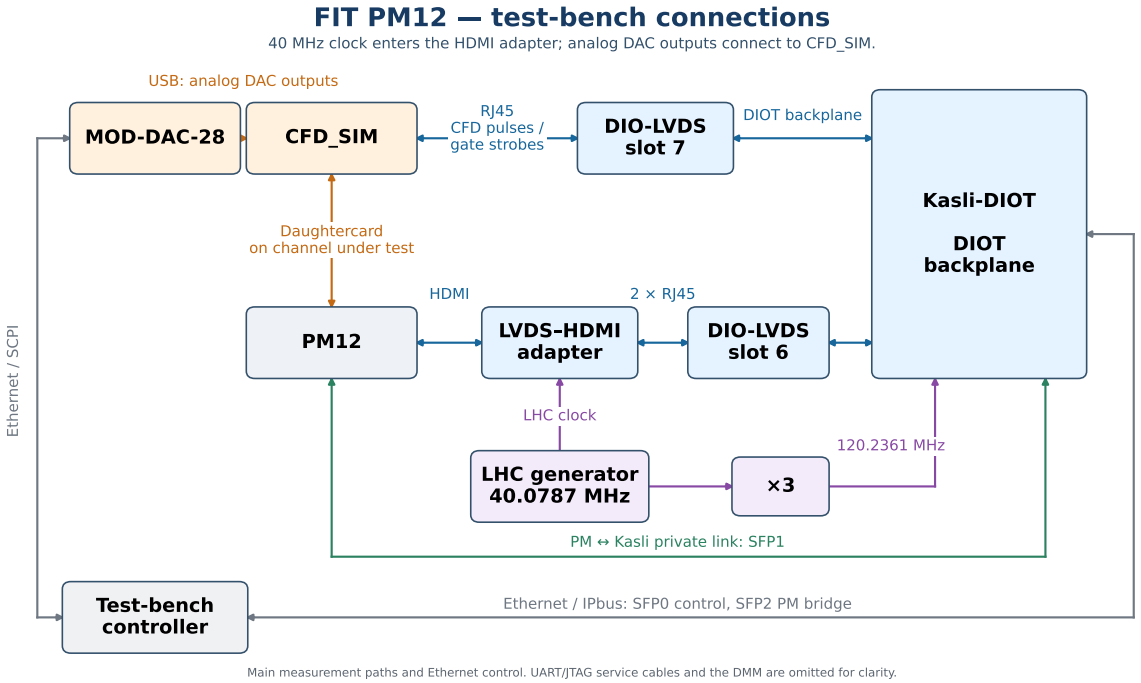
Measurements from multiple dates/configurations. No factory DNA was recorded at acquisition. Not one acceptance run and not automatically bound to a newly enrolled board.

Field	Recorded value
Hardware identity	LEGACY-SN1803-UNBOUND
MCU EEPROM label	1803
Evidence mode	HISTORICAL
Merged batches	1
Analog channels measured	12
Final acceptance	NOT RELEASED — requires complete required coverage, frozen limits and reviewer disposition
Missing channels	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11

Canonical new board label: PM12-K7-<full FPGA factory DNA>. Preserve leading zeroes. JTAG IDCODE identifies chip type, not the individual device. EEPROM serial and debugger cable serial are secondary labels. Replacing the FPGA changes this hardware identity and requires an explicit board genealogy record.

Historical evidence predating factory-DNA reads is never assigned to a newly enrolled DNA automatically. No chip or EEPROM is programmed to assign this number; the printed asset label and registry use the hardware value.

2 Bench inventory and wiring



User-reviewed bench wiring, updated2026-11-08; not a pin-level schematic. Historical wiring and configuration are retained with each acquisition snapshot.

Connection	Physical assignment / observation
Analog stimulus	MOD-DAC-28 analog USB outputs feed CFD_SIM, then the selected PM12 ADC input
HDMI cabling	PM12 HDMI to LVDS-HDMI adapter, then2RJ45 cables to DIO-LVDS slot6 and Kasli DIOT backplane
PM data LVDS via HDMI	Kasli slot6 channels0..3 are inputs; observed PM lane order3,1,2,0
Shared HDMI SPI	Slot6:SELECT4,CLOCK5,MISO6,MOSI7; only4/5/7 may drive; mask0x00b0
CFD cabling and phase observation	CFD_SIM RJ45 to DIO-LVDS slot7 and Kasli backplane; channel0:CFD stimulus; phase RX8/9:channels1/3
PM SFP	Private cable to Kasli SFP1, line rate1.202361Gb/s; not standard switch-facing1000BASE-X
Kasli Ethernet	SFP0:control IPbus; SFP2:optical PM bridge to LAN; SFP3 unused
Common reference	40.0787MHz enters the LVDS-HDMI adapter and reaches PM12 via HDMI; x3 gives Kasli120.2361MHz. Stated settings, no frequency-counter calibration

Role	Module	Identity
DUT	FIT PM12	1803
Timing and readout	Kasli-DIOT	00001
LVDS stimulus/capture	DIO-LVDS RJ45 slot 6	NOT READ
LVDS stimulus/capture	DIO-LVDS RJ45 slot 7	NOT READ
Analog stimulus	MOD-DAC-28	ESP32-MAC-6825DD4C49E4

Role	Module	Identity
DC voltage measurement	Keithley DMM7510	04679690
Analog adapter	CFD_SIM channel 12	NOT READ
Clock reference	External LHC generator	NOT READ

Photographs supplement, but never replace, hardware identity. Each photo is associated with the identified module and its file hash. Missing photos or unread device IDs are left explicit.

FIT PM12

Photograph not supplied
DUT

Kasli-DIOT

Photograph not supplied
Timing and readout

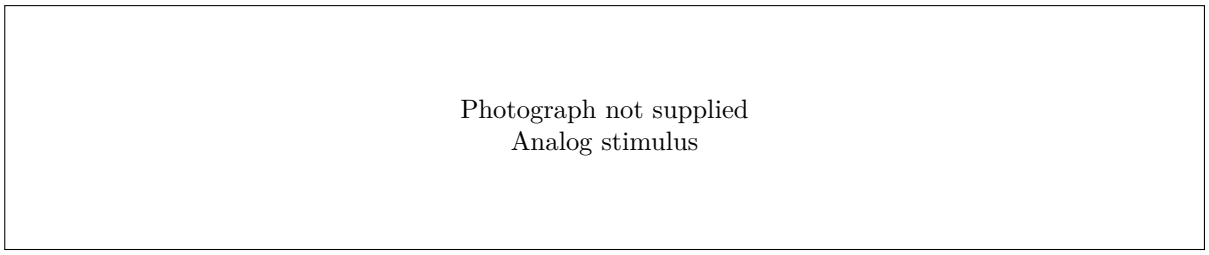
DIO-LVDS RJ45 slot 6

Photograph not supplied
LVDS stimulus/capture

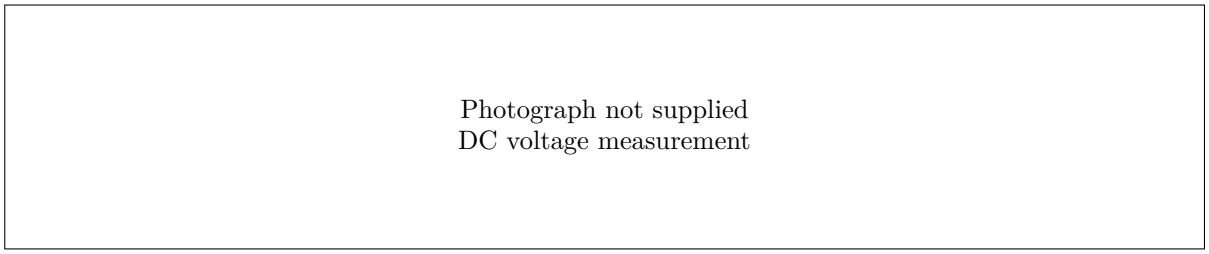
DIO-LVDS RJ45 slot 7

Photograph not supplied
LVDS stimulus/capture

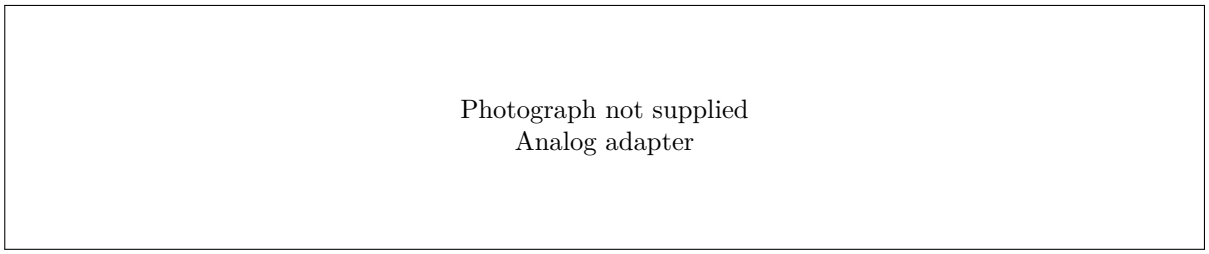
MOD-DAC-28



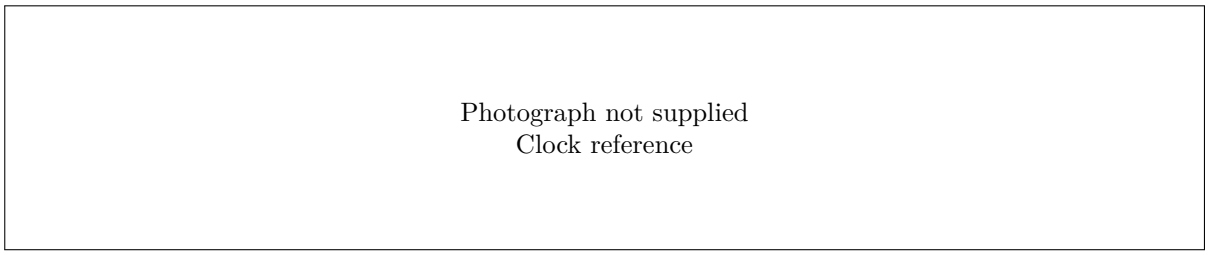
Keithley DMM7510



CFD_SIM channel 12



External LHC generator



3 Test coverage and acceptance rules

Test	Result	Scope / deviation
JTAG chain / 168 static links	HISTORICAL PASS	2026-10-13; exact reduced measured logs retained
CPU–FPGA SPI / peripherals	HISTORICAL PASS	2026-10-15; original diagnostic image
Configuration flash read / ownership	HISTORICAL PASS	Read-only ownership sequence; stored-image boot NOT CONFIRMED
Power startup / timeout fault injection	NOT RUN	Static PGOOD cannot substitute for fault testing
GTX PRBS / Ethernet loopback	HISTORICAL PASS	Old exact 40 MHz setup, not current LHC private link
I2C mux / power monitors	HISTORICAL PASS	2026-10-14; uncalibrated currents, missing P3V8 shunt
HDMI / digital ADC / TDC	HISTORICAL PASS	2026-11-06; all digital ADC channels, stimulated TDC only channel 12
CFD monitor and B/C/D DAC curves	HISTORICAL	Channel 12 only; restored-code correction retained
Gate phase and DAC-A	HISTORICAL PASS	Channel 12,20 settings; nominal sample time units
ADC transfer / adjacent bits	HISTORICAL PASS	2026-11-07 channel 12, A/B; 257 levels, raw revalidated

PASS requires the configured process to finish successfully AND every declared result check to match. NOT_RUN, BLOCKED, ERROR and INCONCLUSIVE cannot become PASS. Failed cleanup is a failure. Repeated runs do not overwrite earlier evidence.

4 Test procedure and criteria

4.1 Hardware identity and PGOOD

Read factory FPGA DNA and MCU EEPROM serial under leases. Match enrolled UID, integrated build and stable physical PGOOD.

Limits: MCU EEPROM serial and FPGA IDCODE are not unique factory identity. CPU factory signature not read.

4.2 JTAG chain and168 static CPLD–FPGA links

Validate13 IDCODEs. Program checksum-verified link diagnostic into FPGA SRAM only.338 walking-one/zero vectors in each of gateware and independent boundary-scan receive paths. Restore checksum-verified integrated image, initialize private link, verify build and repeat HDMI/ADC/TDC functional tests.

Limits: Static connectivity, not at-speed margin or analog quality. Expected SYS reset from deliberate FPGA SRAM reconfiguration; never reset or program MCU/CPLD/flash.

4.3 GTX internal PRBS, MAC loopback and external link

Exercise GTX PRBS and MAC loopback with negative controls; restore integrated image. External PM↔Kasli private link checked separately by integrated suite.

Limits: Legacy1000BASE-X image expects old exact40MHz clock. Requalify with present40.0787MHz; old PASS is historical only.

4.4 I2C mux address inventory and bus topology

Single-branch scans through both muxes, address readback and decoded monitor IDs; preserve current/voltage samples.

Limits: Legacy runner programs older FPGA and restores obsolete baseline. MCU telemetry is integrated, exhaustive topology adapter still requires migration.

4.5 FPGA configuration flash ownership and read-only access

Verify ownership guard, PROGRAM_B hold, JEDEC and bounded READ03, bus release, reload integrated SRAM image.

Limits: Destructive to volatile FPGA state. Qualified current-image restoration/recovery adapter required; stored-image boot remains unconfirmed.

4.6 Power sequencing and timeout shutdown

Observe EN-to-PG rise and timeout shutdown without replacing original MCU protection; record rails and fault state.

Limits: Fault injection and power-cycle qualification NOT_RUN; static PGOOD is not equivalent.

4.7 CPU peripherals, SPI walking patterns and power telemetry

Read PING, GPIO, temperature ID, three TDC chip IDs, PLL, FPGA signature, 32 walking patterns with scratch restoration, INA and POWER. Recheck PGOOD before sensor access.

Limits: No power cycling or timeout fault injection. Unfitted P3V8 shunt: voltage only; no current acceptance. Sensor numeric limits require a frozen acceptance specification.

4.8 HDMI, all digital ADC paths and stimulated TDC

Four HDMI lanes: walking discovery and slow/fast PRBS, empty frames; SPI signature reads only. Capture both ADC banks, synthetic and physical, all12 channels. Quiet/stimulated/quiet TDC:64 pulses to configured channel. Check uptime.

Limits: TDC stimulus only on the physically connected channel. No calibrated ADC analog acceptance for unconnected channels. HDMI SPI writes disabled: bus shared with MCU.

4.9 Gate delay versus internal DAC-A

Establish explicit DAC-A0000 baseline;20 code points0..65535; common trigger and ISERDES phase buffers; restore0000 and verify phase return. Save every raw capture.

Limits: Sampled centroid, nominal sample metadata; no calibrated absolute phase/jitter claim. Original20MHz CPLD gate divider required.

4.10 ADC differential transfer and adjacent data-bit coverage

SCPI DAC differential-1.2..+1.2V, commonmode1V,257 points, ADC A/B. Decode4096-word buffers at STR edges, verify tags/generations;12 bits and11 neighbor pairs, all4 states with sufficient population. Restore1V/1V.

Limits: Requested voltage, no calibrated gain/DNL. Per-pin0..3.3V ceiling. Functional persistent-tie test, not resistance/pin-order/timing-margin proof.

4.11 CFD analog monitor DC levels

Read all8 monitor nodes with DMM7510, correct divider ratios, leave all shared monitor muxes inhibited.

Limits: Original script is channel12-specific. Parameterized multi-module inhibition and DMM/wiring qualification required.

4.12 Internal DAC B/C/D transfer curves

65 code settings perDAC, three DMM readings per setting, corrected analog-node voltage and verified explicit-code restoration.

Limits: Original script is channel12-specific. Per-channel restore codes, shared mux inhibition and DMM wiring must be configured.

5 Digital link and peripheral evidence

5.1 MCU / SPI

Recorded field	Result
scope	Physical MCU SPI with original slave protocol, passive FPGA I2C pins
rows	18 entries; see archived JSON/raw files
mcu_source	f9869c9
fpga_magic	504d5a02
nonvolatile_writes	False
fpga_load	PASS
spi_test	PASS10 runs,320 walking-bit patterns with readback and scratch restoration
restored_image	Matching passive SPI diagnostic left in SRAM

5.2 Flash ownership

Recorded field	Result
scope	Physical config bus ownership/read-only flash/reset-release; no flash write/erase
rows	18 entries; see archived JSON/raw files
jedec_id	016017
flash_ownership_read_release	PASS
boot_done_within2s	False
boot_validation	DONE did not assert; stored image unknown, not a successful FPGA boot

5.3 Ethernet / PRBS

Recorded field	Result
board	PM12
serial	210249A73E43
external_sfp	NOT_RUN
persistent_writes	False
status	PASS
bitstream_sha256	fb6e39d8375d46dd01fde00d22a6075bf78756666cb79a8cc96f235622a5
restore_bitstream_sha256	c9fbd02768ce12750dc68b8bb24c34403027a65f9e002661e29bb3d2e2d
samples	12 entries; see archived JSON/raw files
checks.initial_mac_clocks	True
checks.initial_mac	True
checks.payload_recovery_clocks	True
checks.payload_recovery	True
checks.prbs_clocks	True
checks.prbs	True
checks.prbs_recovery_clocks	True
checks.prbs_recovery	True
checks.final_mac_clocks	True
checks.final_mac	True
checks.payload_negative_control	True
checks.prbs_negative_control	True
windows.initial_mac.tick	360504425
windows.initial_mac.refcount	300420354

Recorded field	Result
windows.initial_mac.txframes	88576
windows.initial_mac.rxframes	88576
windows.initial_mac.badframes	0
windows.initial_mac.prbs_error_samples	0
windows.initial_mac.rxbytes	5314560
windows.initial_mac.sys_mhz	120.00813082556591

5.4 I2C topology

Recorded field	Result
date	2026-10-14
digital_pass	True
physical_restore_verified	True
fpga_sha256	b26bdccb44ff0d323a58637efd3cee59a98752afdf432d392f4aa47a1a3d9
nonvolatile_writes	False
mux_addresses	['0x70', '0x71']
mux_selection_readback_cases	16
reset_test	True
automatic_line_release_test	True
downstream_scan_address_range	0x08..0x77 excluding upstream mux addresses
confirmed_monitors	3
afe_device_ack_count	0
warnings	2 entries; see archived JSON/raw files
coverage_limits	['AFE routes yielded no downstream ACK; physical connectivity of unloaded branches is not proven', 'No sensor configuration or power-controller writes', 'Voltage/current calibration not validated', 'Slow host-driven I2C over UDP; high-speed I2C electrical margins not measured']
bus_voltages_by_monitor.4	[1.0, 1.2, 1.792]
bus_voltages_by_monitor.5	[1.456, 3.336, 1.784]
bus_voltages_by_monitor.6	[1.008, 5.5040000000000004, 0.0]

5.5 Power rail observations

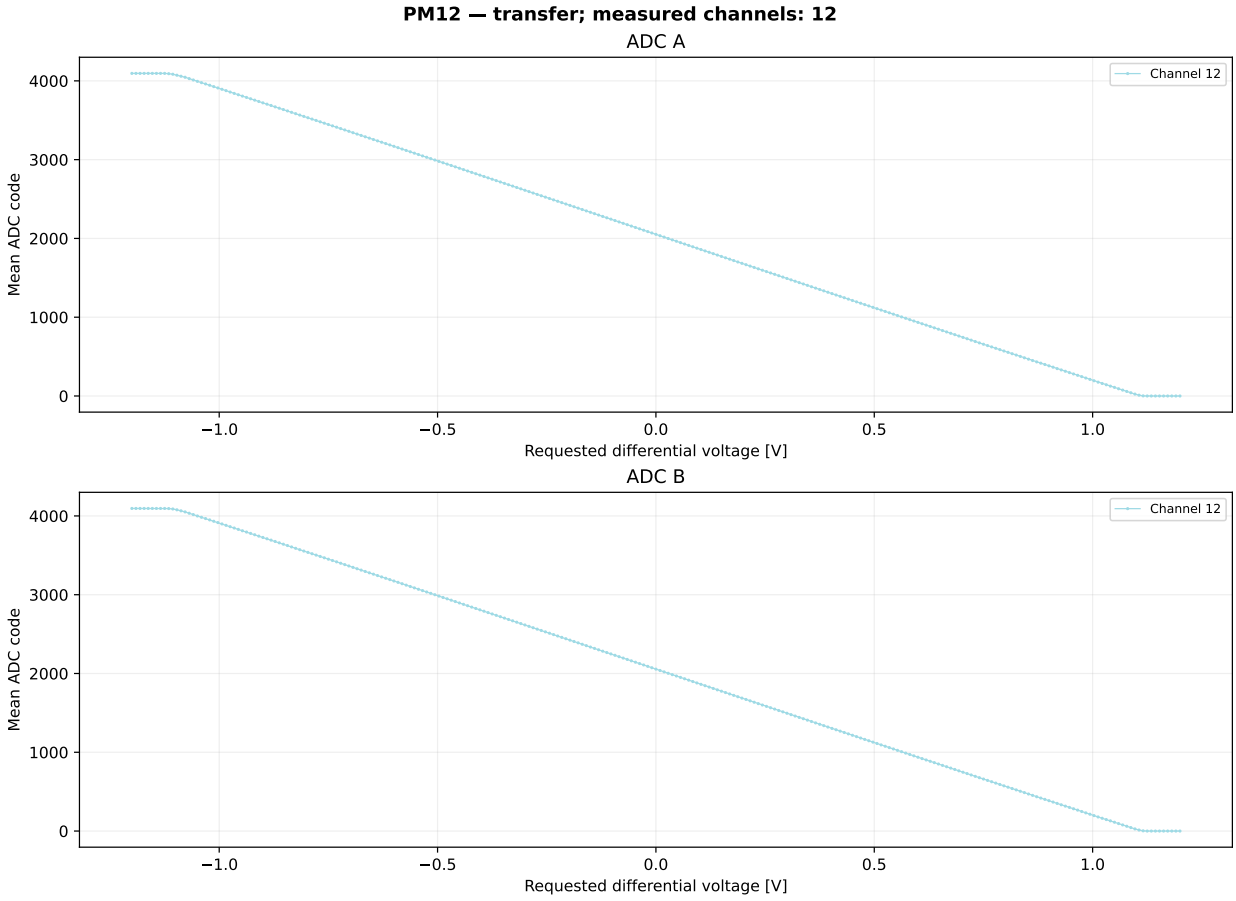
Rail	Bus [V]	Current [A]	Note
P1V0_LDO	1.0	0.1276	Uncalibrated; inspect shunt/clipping metadata
P1V2_LDO	1.2	0.092	Uncalibrated; inspect shunt/clipping metadata
P1V8_LDO	1.792	0.004	Uncalibrated; inspect shunt/clipping metadata
P3V8	1.48	None	Uncalibrated; inspect shunt/clipping metadata
P3V3	3.336	4.920000000000001	Uncalibrated; inspect shunt/clipping metadata
P1V8	1.784	0.36000000000000004	Uncalibrated; inspect shunt/clipping metadata
P1V0	1.008	0.10800000000000001	Uncalibrated; inspect shunt/clipping metadata
P5V5	5.5040000000000004	0.08	Uncalibrated; inspect shunt/clipping metadata

Rail	Bus [V]	Current [A]	Note
TP1/TP2	0.0	None	Uncalibrated; inspect shunt/clipping metadata

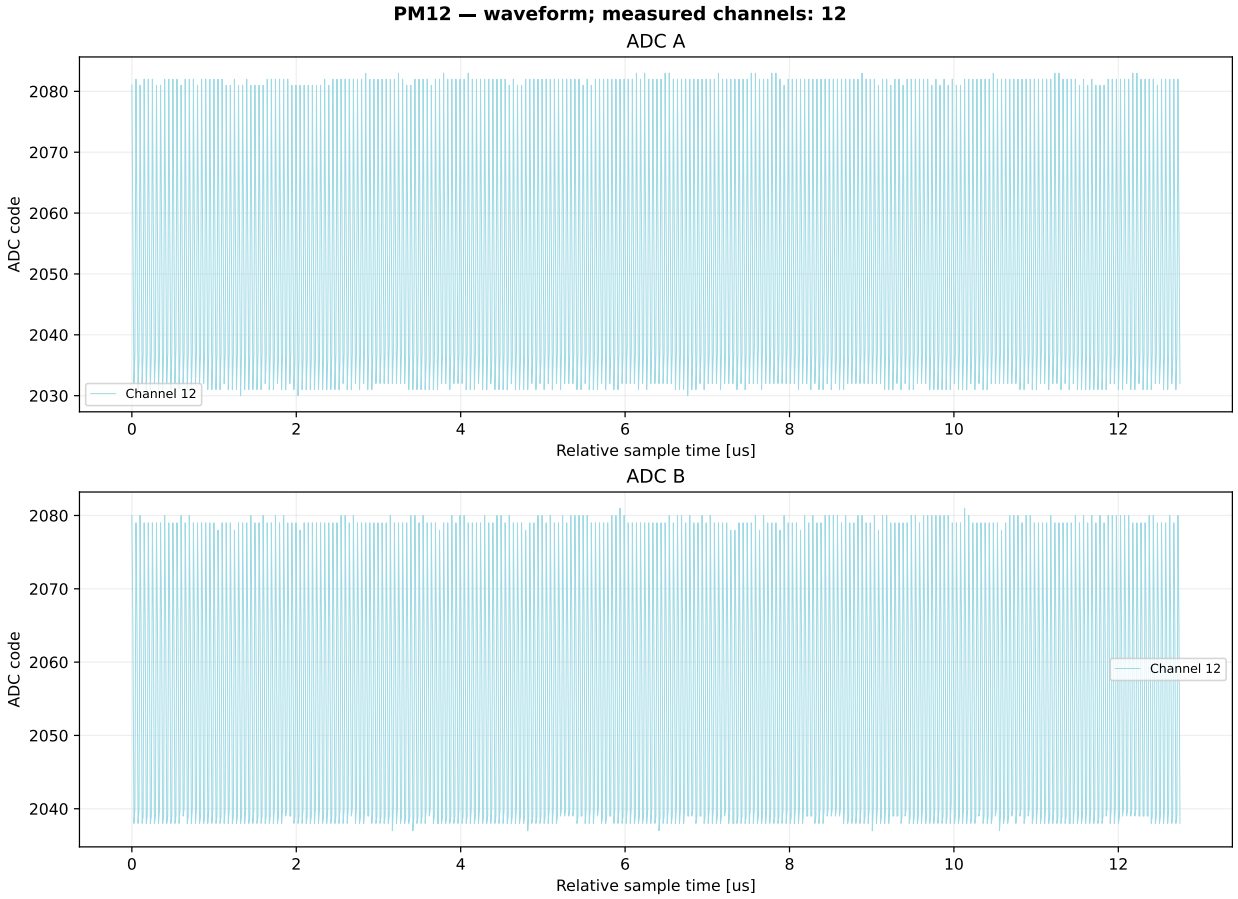
P3V8 has no fitted shunt: current is not an accepted measurement. Original clipping observations are preserved in raw telemetry and are not interpreted as a rail current.

6 CFD and ADC — channel comparison

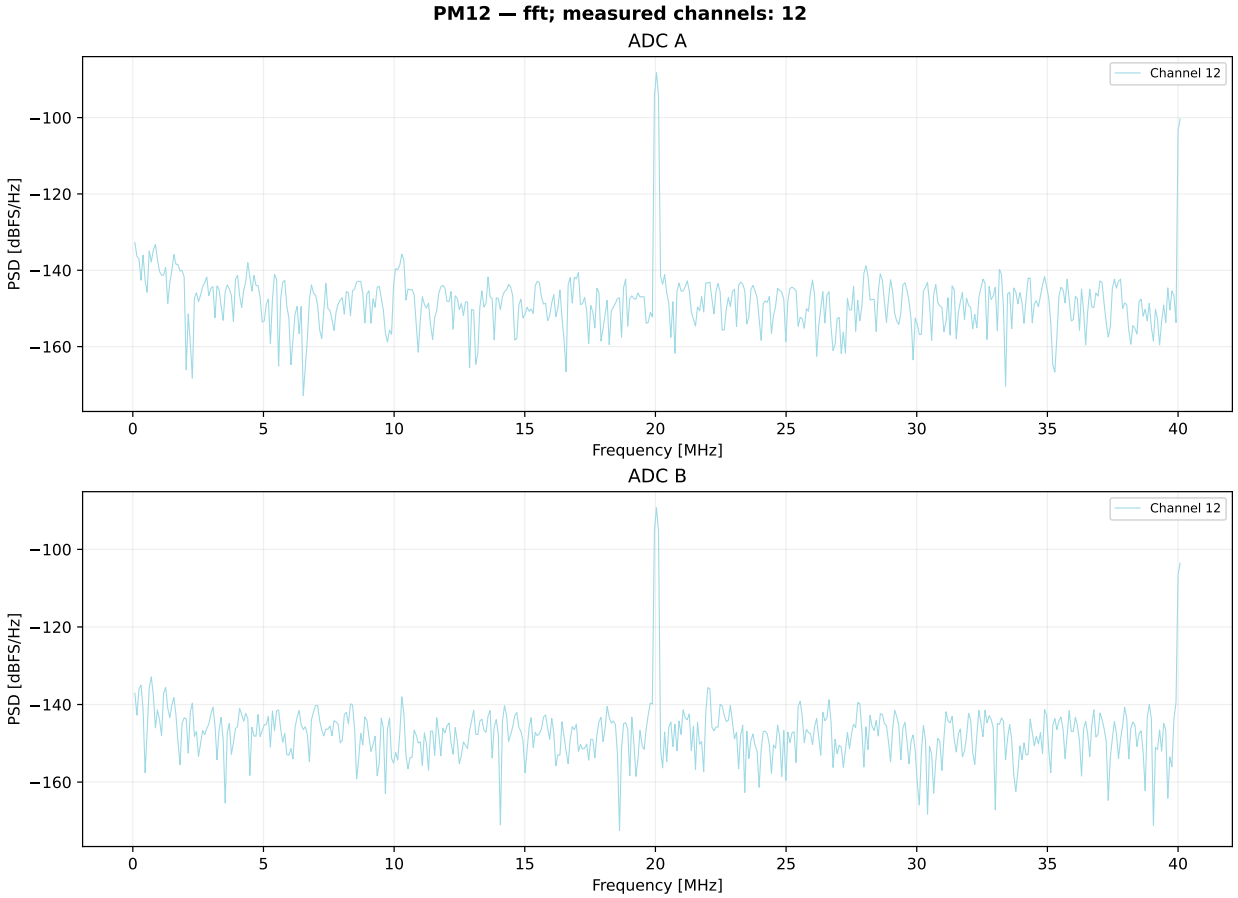
All measured channels share one overlay per observable, with a fixed channel color and common axes. ADC A/B occupy separate panels. No line is invented for an unmeasured channel. The twelve-channel requirement is independent of the number of connected simulators in a batch.



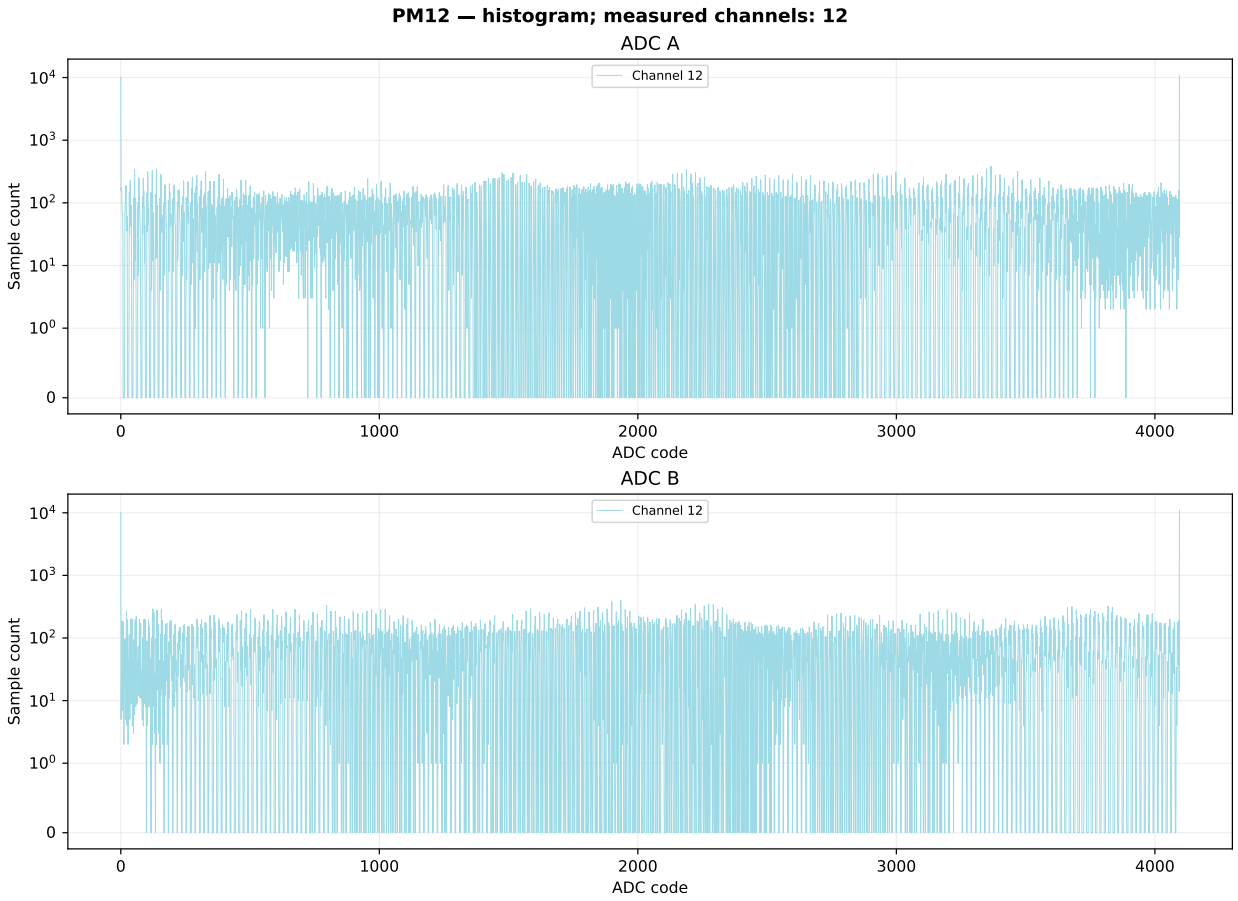
Requested DAC voltage, not a calibrated voltage measurement.



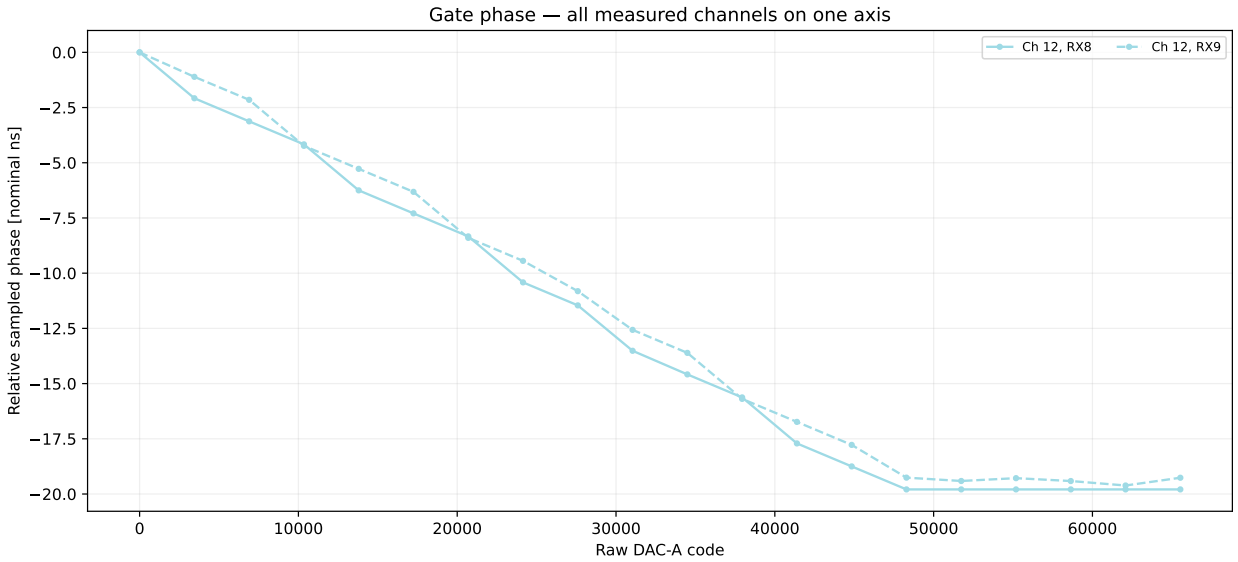
One sample per STR edge; time origins are local to sequential captures. Fs inferred as 80.1574 MHz.



DC removed, Hann window, one-sided PSD. Fs inferred from stated LHC reference, not measured by a frequency counter. No ENOB/SINAD acceptance from a DC capture.



Stepped DC code density; unvisited codes are not automatically ADC missing-code/DNL faults.



Nominal sampled-centroid units; original 20 MHz gate divider. Curves use the same reported sample metadata.

Channel	ADC	Line exercise	Visited codes	Interior samples
1	A	NOT MEASURED	-	-
1	B	NOT MEASURED	-	-
2	A	NOT MEASURED	-	-
2	B	NOT MEASURED	-	-
3	A	NOT MEASURED	-	-
3	B	NOT MEASURED	-	-
4	A	NOT MEASURED	-	-
4	B	NOT MEASURED	-	-

Channel	ADC	Line exercise	Visited codes	Interior samples
5	A	NOT MEASURED	-	-
5	B	NOT MEASURED	-	-
6	A	NOT MEASURED	-	-
6	B	NOT MEASURED	-	-
7	A	NOT MEASURED	-	-
7	B	NOT MEASURED	-	-
8	A	NOT MEASURED	-	-
8	B	NOT MEASURED	-	-
9	A	NOT MEASURED	-	-
9	B	NOT MEASURED	-	-
10	A	NOT MEASURED	-	-
10	B	NOT MEASURED	-	-
11	A	NOT MEASURED	-	-
11	B	NOT MEASURED	-	-
12	A	PASS	3129	241856
12	B	PASS	3057	241760

ADC	Channel	Max deviation [codes]	RMS deviation [codes]

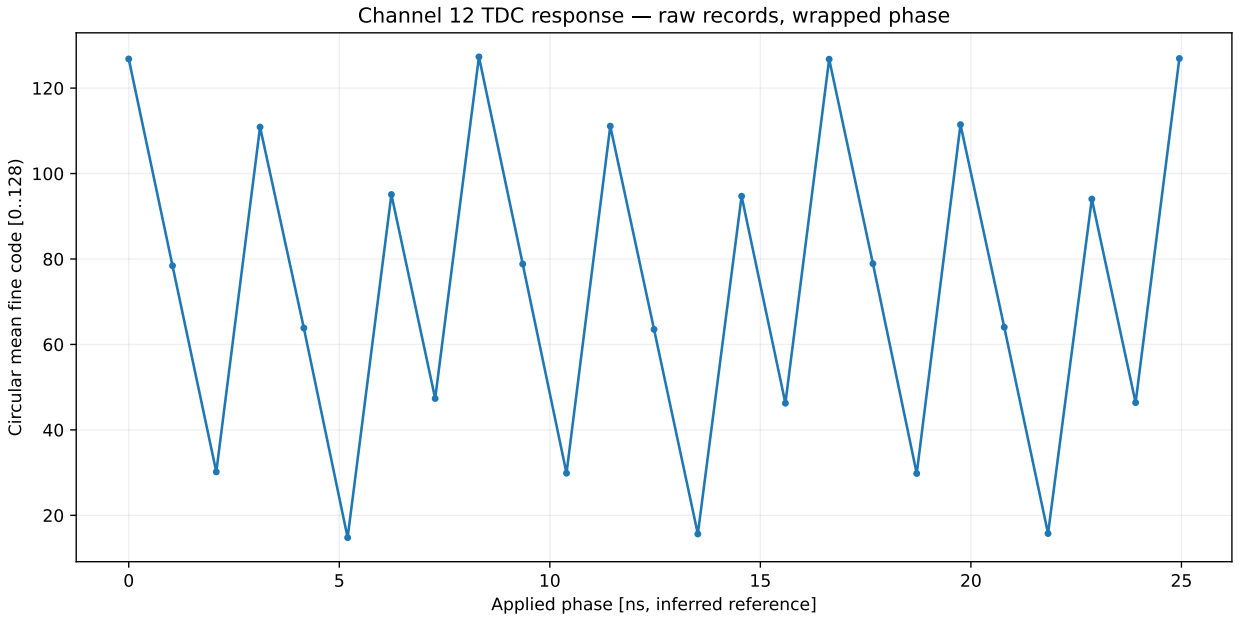
Deviation is relative to the median of measured channels at common requested stimulus settings, without interpolation. With fewer than two channels this comparison is NOT COMPARABLE, rather than zero dispersion. Values are descriptive, not a calibrated gain/error acceptance limit.

Bit decisions exclude rail codes 0/4095. Both values of every bit and all four states of each neighboring pair must be populated: at least 64 samples and 2% of all interior samples per state, plus at least 8 samples at each of 3 distinct settings. At least 16 settings, 4096 interior samples and 3000-code mean span are required for PASS.

ADC strobe validation uses the4x sampled clock model:1023 or1024 rising and falling edges, intervals3/4/5 only, cumulative rising and falling phase spans at most one sample, and pulse widths1..3 samples. The ADC word selected immediately before STR must equal its preceding sample. EDGE QUANTIZED is reported separately from ALIGNED; neither is a calibrated jitter result. A strict all4-interval check is a separate phase-alignment diagnostic.

This excludes sampled persistent stuck/tied signatures under these conditions. It does not establish electrical resistance, bit order, timing margin, ADC INL/DNL or full missing-code compliance. A sine histogram is arcsine-shaped; a calibrated continuous ramp is a separate density test.

7 TDC phase response



Recomputed circular means from raw 13-bit TDC records. Response and event counting, not calibrated picosecond accuracy.

Applied slot	Applied delay [ns]	Fine circular mean	Events
24	0	126.828	128
25	1.03962	78.4219	128
26	2.07924	30.164	128
27	3.11886	110.883	128
28	4.15848	63.836	128
29	5.19811	14.7735	128
30	6.23773	95.0938	128
31	7.27735	47.3438	128
32	8.31697	127.312	128
33	9.35659	78.8204	128
34	10.3962	29.8593	128
35	11.4358	111.102	128
36	12.4755	63.5234	128
37	13.5151	15.6329	128
38	14.5547	94.6953	128
39	15.5943	46.2422	128
40	16.6339	126.766	128
41	17.6736	78.8984	128
42	18.7132	29.789	128
43	19.7528	111.453	128
44	20.7924	64.0389	128
45	21.832	15.7422	128
46	22.8717	94.0311	128
47	23.9113	46.375	128
48	24.9509	126.938	128

8 Internal CFD DAC B/C/D characteristics

Physical channel 12 only. First attempted restoration using computed STORED parameters was rejected; explicit active-code 0000 restoration then reproduced initial voltages. The original failed attempt remains in the source evidence.

DAC	Signal	Start [V]	End [V]
B	DRV_OFFSET	1.043846	1.003721
C	CFD_OFFSET	1.034097	1.0136713333333334
D	CFD_CMP_OFFSET	0.004514859666666667	1.0626073333333335

9 Static connection results

Results follow the static connection procedure in the test-procedure chapter. All rows below share one discovery and schematic revision; at-speed and analog tests are documented separately.

CPLD endpoints are physical channel / package pin. Each result includes walking-one, walking-zero, all-zero/all-one baselines and agreement with the schematic. Vector indices, boundary cells and full per-vector observations remain in the archived JSON/logs.

Signal	FPGA ball	Expected CPLD	Gateware CPLD	Boundary CPLD	Gateware	Boundary
DI1[0]	T7	1 / 99	1 / 99	1 / 99	PASS	PASS
DI1[1]	U7	1 / 97	1 / 97	1 / 97	PASS	PASS
DI1[2]	U6	1 / 96	1 / 96	1 / 96	PASS	PASS
DI1[3]	U5	1 / 95	1 / 95	1 / 95	PASS	PASS
DI1[4]	U4	1 / 94	1 / 94	1 / 94	PASS	PASS
DI1[5]	U2	1 / 93	1 / 93	1 / 93	PASS	PASS
DI1[6]	U1	1 / 92	1 / 92	1 / 92	PASS	PASS
DI1[7]	V6	1 / 91	1 / 91	1 / 91	PASS	PASS
DI1[8]	V4	1 / 90	1 / 90	1 / 90	PASS	PASS
DI1[9]	W4	1 / 89	1 / 89	1 / 89	PASS	PASS
DI1[10]	V2	1 / 87	1 / 87	1 / 87	PASS	PASS
DI1[11]	V1	1 / 86	1 / 86	1 / 86	PASS	PASS
DI1[12]	W6	1 / 85	1 / 85	1 / 85	PASS	PASS
STR1	W5	1 / 82	1 / 82	1 / 82	PASS	PASS
DI2[0]	AF4	2 / 99	2 / 99	2 / 99	PASS	PASS
DI2[1]	AD5	2 / 97	2 / 97	2 / 97	PASS	PASS
DI2[2]	AE5	2 / 96	2 / 96	2 / 96	PASS	PASS
DI2[3]	AF5	2 / 95	2 / 95	2 / 95	PASS	PASS
DI2[4]	AB6	2 / 94	2 / 94	2 / 94	PASS	PASS
DI2[5]	AC6	2 / 93	2 / 93	2 / 93	PASS	PASS
DI2[6]	AD6	2 / 92	2 / 92	2 / 92	PASS	PASS
DI2[7]	AE6	2 / 91	2 / 91	2 / 91	PASS	PASS
DI2[8]	U9	2 / 90	2 / 90	2 / 90	PASS	PASS
DI2[9]	V7	2 / 89	2 / 89	2 / 89	PASS	PASS
DI2[10]	V8	2 / 87	2 / 87	2 / 87	PASS	PASS
DI2[11]	Y7	2 / 86	2 / 86	2 / 86	PASS	PASS
DI2[12]	Y8	2 / 85	2 / 85	2 / 85	PASS	PASS
STR2	AA7	2 / 82	2 / 82	2 / 82	PASS	PASS
DI3[0]	AA8	3 / 99	3 / 99	3 / 99	PASS	PASS
DI3[1]	AD11	3 / 97	3 / 97	3 / 97	PASS	PASS
DI3[2]	AE11	3 / 96	3 / 96	3 / 96	PASS	PASS
DI3[3]	V12	3 / 95	3 / 95	3 / 95	PASS	PASS
DI3[4]	Y12	3 / 94	3 / 94	3 / 94	PASS	PASS
DI3[5]	Y13	3 / 93	3 / 93	3 / 93	PASS	PASS
DI3[6]	AA12	3 / 92	3 / 92	3 / 92	PASS	PASS
DI3[7]	AA13	3 / 91	3 / 91	3 / 91	PASS	PASS
DI3[8]	V13	3 / 90	3 / 90	3 / 90	PASS	PASS
DI3[9]	W13	3 / 89	3 / 89	3 / 89	PASS	PASS
DI3[10]	V14	3 / 87	3 / 87	3 / 87	PASS	PASS
DI3[11]	W14	3 / 86	3 / 86	3 / 86	PASS	PASS
DI3[12]	AA14	3 / 85	3 / 85	3 / 85	PASS	PASS
STR3	AA15	3 / 82	3 / 82	3 / 82	PASS	PASS
DI4[0]	V18	4 / 99	4 / 99	4 / 99	PASS	PASS
DI4[1]	W18	4 / 97	4 / 97	4 / 97	PASS	PASS
DI4[2]	AA19	4 / 96	4 / 96	4 / 96	PASS	PASS
DI4[3]	AC19	4 / 95	4 / 95	4 / 95	PASS	PASS
DI4[4]	AE18	4 / 94	4 / 94	4 / 94	PASS	PASS
DI4[5]	AF18	4 / 93	4 / 93	4 / 93	PASS	PASS
DI4[6]	AB20	4 / 92	4 / 92	4 / 92	PASS	PASS
DI4[7]	V19	4 / 91	4 / 91	4 / 91	PASS	PASS
DI4[8]	W19	4 / 90	4 / 90	4 / 90	PASS	PASS
DI4[9]	AB19	4 / 89	4 / 89	4 / 89	PASS	PASS
DI4[10]	AD19	4 / 87	4 / 87	4 / 87	PASS	PASS
DI4[11]	AF19	4 / 86	4 / 86	4 / 86	PASS	PASS

Signal	FPGA ball	Expected CPLD	Gateware CPLD	Boundary CPLD	Gateware	Boundary
DI4[12]	AA20	4 / 85	4 / 85	4 / 85	PASS	PASS
STR4	AD20	4 / 82	4 / 82	4 / 82	PASS	PASS
DI5[0]	AE20	5 / 99	5 / 99	5 / 99	PASS	PASS
DI5[1]	AF20	5 / 97	5 / 97	5 / 97	PASS	PASS
DI5[2]	AE21	5 / 96	5 / 96	5 / 96	PASS	PASS
DI5[3]	AF22	5 / 95	5 / 95	5 / 95	PASS	PASS
DI5[4]	AE22	5 / 94	5 / 94	5 / 94	PASS	PASS
DI5[5]	AF23	5 / 93	5 / 93	5 / 93	PASS	PASS
DI5[6]	AE23	5 / 92	5 / 92	5 / 92	PASS	PASS
DI5[7]	AF24	5 / 91	5 / 91	5 / 91	PASS	PASS
DI5[8]	AF25	5 / 90	5 / 90	5 / 90	PASS	PASS
DI5[9]	AE25	5 / 89	5 / 89	5 / 89	PASS	PASS
DI5[10]	AE26	5 / 87	5 / 87	5 / 87	PASS	PASS
DI5[11]	AD21	5 / 86	5 / 86	5 / 86	PASS	PASS
DI5[12]	AD23	5 / 85	5 / 85	5 / 85	PASS	PASS
STR5	AB21	5 / 82	5 / 82	5 / 82	PASS	PASS
DI6[0]	AD25	6 / 99	6 / 99	6 / 99	PASS	PASS
DI6[1]	AD26	6 / 97	6 / 97	6 / 97	PASS	PASS
DI6[2]	AC21	6 / 96	6 / 96	6 / 96	PASS	PASS
DI6[3]	AC22	6 / 95	6 / 95	6 / 95	PASS	PASS
DI6[4]	AC23	6 / 94	6 / 94	6 / 94	PASS	PASS
DI6[5]	AD24	6 / 93	6 / 93	6 / 93	PASS	PASS
DI6[6]	AC26	6 / 92	6 / 92	6 / 92	PASS	PASS
DI6[7]	AC24	6 / 91	6 / 91	6 / 91	PASS	PASS
DI6[8]	AB22	6 / 90	6 / 90	6 / 90	PASS	PASS
DI6[9]	AB24	6 / 89	6 / 89	6 / 89	PASS	PASS
DI6[10]	AB25	6 / 87	6 / 87	6 / 87	PASS	PASS
DI6[11]	AB26	6 / 86	6 / 86	6 / 86	PASS	PASS
DI6[12]	AA22	6 / 85	6 / 85	6 / 85	PASS	PASS
STR6	AA23	6 / 82	6 / 82	6 / 82	PASS	PASS
DI7[0]	AA24	7 / 99	7 / 99	7 / 99	PASS	PASS
DI7[1]	AA25	7 / 97	7 / 97	7 / 97	PASS	PASS
DI7[2]	Y20	7 / 96	7 / 96	7 / 96	PASS	PASS
DI7[3]	Y21	7 / 95	7 / 95	7 / 95	PASS	PASS
DI7[4]	Y22	7 / 94	7 / 94	7 / 94	PASS	PASS
DI7[5]	Y23	7 / 93	7 / 93	7 / 93	PASS	PASS
DI7[6]	Y25	7 / 92	7 / 92	7 / 92	PASS	PASS
DI7[7]	Y26	7 / 91	7 / 91	7 / 91	PASS	PASS
DI7[8]	W20	7 / 90	7 / 90	7 / 90	PASS	PASS
DI7[9]	W21	7 / 89	7 / 89	7 / 89	PASS	PASS
DI7[10]	W23	7 / 87	7 / 87	7 / 87	PASS	PASS
DI7[11]	W24	7 / 86	7 / 86	7 / 86	PASS	PASS
DI7[12]	W25	7 / 85	7 / 85	7 / 85	PASS	PASS
STR7	W26	7 / 82	7 / 82	7 / 82	PASS	PASS
DI8[0]	V21	8 / 99	8 / 99	8 / 99	PASS	PASS
DI8[1]	V22	8 / 97	8 / 97	8 / 97	PASS	PASS
DI8[2]	V23	8 / 96	8 / 96	8 / 96	PASS	PASS
DI8[3]	V24	8 / 95	8 / 95	8 / 95	PASS	PASS
DI8[4]	V26	8 / 94	8 / 94	8 / 94	PASS	PASS
DI8[5]	U21	8 / 93	8 / 93	8 / 93	PASS	PASS
DI8[6]	U22	8 / 92	8 / 92	8 / 92	PASS	PASS
DI8[7]	U24	8 / 91	8 / 91	8 / 91	PASS	PASS
DI8[8]	U25	8 / 90	8 / 90	8 / 90	PASS	PASS
DI8[9]	U26	8 / 89	8 / 89	8 / 89	PASS	PASS
DI8[10]	U16	8 / 87	8 / 87	8 / 87	PASS	PASS
DI8[11]	U17	8 / 86	8 / 86	8 / 86	PASS	PASS
DI8[12]	U19	8 / 85	8 / 85	8 / 85	PASS	PASS
STR8	U20	8 / 82	8 / 82	8 / 82	PASS	PASS
DI9[0]	T17	9 / 99	9 / 99	9 / 99	PASS	PASS
DI9[1]	T18	9 / 97	9 / 97	9 / 97	PASS	PASS
DI9[2]	T19	9 / 96	9 / 96	9 / 96	PASS	PASS
DI9[3]	T20	9 / 95	9 / 95	9 / 95	PASS	PASS
DI9[4]	T22	9 / 94	9 / 94	9 / 94	PASS	PASS

Signal	FPGA ball	Expected CPLD	Gateware CPLD	Boundary CPLD	Gateware	Boundary
DI9[5]	T23	9 / 93	9 / 93	9 / 93	PASS	PASS
DI9[6]	T24	9 / 92	9 / 92	9 / 92	PASS	PASS
DI9[7]	T25	9 / 91	9 / 91	9 / 91	PASS	PASS
DI9[8]	R16	9 / 90	9 / 90	9 / 90	PASS	PASS
DI9[9]	R17	9 / 89	9 / 89	9 / 89	PASS	PASS
DI9[10]	R18	9 / 87	9 / 87	9 / 87	PASS	PASS
DI9[11]	R20	9 / 86	9 / 86	9 / 86	PASS	PASS
DI9[12]	R21	9 / 85	9 / 85	9 / 85	PASS	PASS
STR9	R22	9 / 82	9 / 82	9 / 82	PASS	PASS
DI10[0]	R23	10 / 99	10 / 99	10 / 99	PASS	PASS
DI10[1]	R25	10 / 97	10 / 97	10 / 97	PASS	PASS
DI10[2]	R26	10 / 96	10 / 96	10 / 96	PASS	PASS
DI10[3]	P16	10 / 95	10 / 95	10 / 95	PASS	PASS
DI10[4]	P18	10 / 94	10 / 94	10 / 94	PASS	PASS
DI10[5]	P19	10 / 93	10 / 93	10 / 93	PASS	PASS
DI10[6]	P20	10 / 92	10 / 92	10 / 92	PASS	PASS
DI10[7]	P21	10 / 91	10 / 91	10 / 91	PASS	PASS
DI10[8]	P23	10 / 90	10 / 90	10 / 90	PASS	PASS
DI10[9]	P24	10 / 89	10 / 89	10 / 89	PASS	PASS
DI10[10]	P25	10 / 87	10 / 87	10 / 87	PASS	PASS
DI10[11]	P26	10 / 86	10 / 86	10 / 86	PASS	PASS
DI10[12]	N16	10 / 85	10 / 85	10 / 85	PASS	PASS
STR10	N17	10 / 82	10 / 82	10 / 82	PASS	PASS
DI11[0]	N18	11 / 99	11 / 99	11 / 99	PASS	PASS
DI11[1]	N19	11 / 97	11 / 97	11 / 97	PASS	PASS
DI11[2]	N21	11 / 96	11 / 96	11 / 96	PASS	PASS
DI11[3]	N22	11 / 95	11 / 95	11 / 95	PASS	PASS
DI11[4]	N23	11 / 94	11 / 94	11 / 94	PASS	PASS
DI11[5]	N24	11 / 93	11 / 93	11 / 93	PASS	PASS
DI11[6]	N26	11 / 92	11 / 92	11 / 92	PASS	PASS
DI11[7]	M19	11 / 91	11 / 91	11 / 91	PASS	PASS
DI11[8]	M20	11 / 90	11 / 90	11 / 90	PASS	PASS
DI11[9]	M21	11 / 89	11 / 89	11 / 89	PASS	PASS
DI11[10]	M22	11 / 87	11 / 87	11 / 87	PASS	PASS
DI11[11]	M24	11 / 86	11 / 86	11 / 86	PASS	PASS
DI11[12]	M25	11 / 85	11 / 85	11 / 85	PASS	PASS
STR11	M26	11 / 82	11 / 82	11 / 82	PASS	PASS
DI12[0]	L24	12 / 99	12 / 99	12 / 99	PASS	PASS
DI12[1]	L25	12 / 97	12 / 97	12 / 97	PASS	PASS
DI12[2]	K25	12 / 96	12 / 96	12 / 96	PASS	PASS
DI12[3]	K26	12 / 95	12 / 95	12 / 95	PASS	PASS
DI12[4]	M16	12 / 94	12 / 94	12 / 94	PASS	PASS
DI12[5]	M17	12 / 93	12 / 93	12 / 93	PASS	PASS
DI12[6]	L17	12 / 92	12 / 92	12 / 92	PASS	PASS
DI12[7]	L18	12 / 91	12 / 91	12 / 91	PASS	PASS
DI12[8]	L19	12 / 90	12 / 90	12 / 90	PASS	PASS
DI12[9]	L20	12 / 89	12 / 89	12 / 89	PASS	PASS
DI12[10]	K18	12 / 87	12 / 87	12 / 87	PASS	PASS
DI12[11]	K16	12 / 86	12 / 86	12 / 86	PASS	PASS
DI12[12]	K17	12 / 85	12 / 85	12 / 85	PASS	PASS
STR12	J19	12 / 82	12 / 82	12 / 82	PASS	PASS

Common test condition	Recorded value
Discovery vectors per receive path	338
Functional CPLD restoration	True

Schematic SHA256: 8a33ca4d4325eea999347ee7f4c26173cb28356f5cb61e6d38a9852a0513de78

10 Raw data, reproducibility and review

Batch	Mode / original status
00-historical	HISTORICAL / INCOMPLETE

The inputs directory retains complete evidence bundles. files.sha256.json lists every file hash and size. Generated figures retain their offline analysis JSON. Edit report.tex for typography and content.tex for this report; edit the source template for future reports. Rebuilding figures does not contact instruments.

Review checklist: verify factory UID and photographs; confirm pin-level cabling; approve numerical acceptance limits and calibration status; review every failure and deviation; confirm channel coverage and cleanup; retain raw records with this signed PDF.

No result is an assertion that the hardware is free of every possible defect. The evidence defines tested behavior, test conditions and observed limitations. This dossier is not a substitute for the separate CERN-equivalent detector validation bench.